

# PAUL MITCHELL

Riverdale, MD · (413) 204-0493 · [pjmitchell@wm.edu](mailto:pjmitchell@wm.edu) · [linkedin.com/in/paul-mitchell-b1497b30b](https://www.linkedin.com/in/paul-mitchell-b1497b30b) · [github.com/pjmitchell7](https://github.com/pjmitchell7)

## TECHNOLOGY CONSULTING · FEDERAL CLOUD & SYSTEMS ARCHITECTURE

### PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, AI/ML & Systems) and clearable U.S. citizen targeting CGI Federal's Technology Consulting practice. Combines production Linux systems administration (100+ servers), secure cloud automation, and Prometheus/Grafana telemetry with hands-on pre-inference LLM defense proxy engineering and federal compliance readiness (STIX 2.1, ArcSight CEF). Adept in translating stringent federal cybersecurity directives into scalable, production-grade cloud implementations.

### EDUCATION

**College of William & Mary — M.S. in Computer Science (AI/ML & Systems)** Expected Dec 2026 · GPA: 3.95  
**College of William & Mary — B.S. in Computer Science** Graduated Dec 2024  
Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

### TECHNICAL SKILLS & INFRASTRUCTURE

|                                     |                                                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Cloud &amp; Systems:</b>         | AWS (EC2, S3, IAM), Linux (Ubuntu/RHEL), Docker, Kubernetes, Puppet, DNS, SSH, Netplan, Bash, Git   |
| <b>Monitoring &amp; Operations:</b> | Prometheus, Grafana, OpenTelemetry, alert rules, incident triage, log auditing, capacity planning   |
| <b>Programming &amp; Back-End:</b>  | Python, FastAPI, Java, C/C++, SQL (PostgreSQL), YAML, Git/GitLab CI, RESTful APIs, shell automation |
| <b>Security &amp; Compliance:</b>   | Pre-inference AI defense, STIX 2.1, ArcSight CEF, zero-width steganography, threshold calibration   |

### TECHNICAL & PROFESSIONAL EXPERIENCE

**IT Administrator | William & Mary Technology Support Center | Williamsburg, VA** Feb 2025 – Present

- Administer, troubleshoot, and support over 100 Linux instructional compute servers, resolving network connectivity, DNS, routing, account, and application incidents.
- Orchestrated centralized configuration automation via Puppet, standardizing baseline security policies across multi-room labs.
- Built Prometheus and Grafana telemetry dashboards monitoring GPU allocation, compute metrics, and storage health, cutting downtime response times by over 40%.
- Conducted architectural evaluation of multi-user Docker containerization, producing technical trade-off documentation on isolation boundaries.
- Standardized network routing and static IP configurations, maintaining strict configuration control across all production instructional hosts.
- Automated proactive health monitoring daemons in Bash, detecting corrupted filesystem tables and zombie processes before user impact.

**Data Analytics & AI Engineering Extern | Extern (Amazon-Sponsored Project) | Remote** Jun 2025 – Sep 2025

- Engineered reproducible Python and AWS S3 data pipelines to ingest, clean, and validate ~20,000 text records, cutting recurring data prep cycles by 85%.
- Fine-tuned DistilBERT classification models ( $F1 = 0.89$ ) and applied BERTopic clustering to isolate operational turnover factors.
- Trained predictive LightGBM decision models and deployed an interactive Streamlit dashboard delivering real-time metrics for program mentors.
- Implemented dataset versioning using DVC on AWS S3 to ensure reproducibility and audit compliance across multi-team evaluations.

### KEY TECHNICAL & SYSTEMS PROJECTS

**ShadowPrompt: Pre-Inference Defense Proxy for Enterprise LLMs | Python, FastAPI, Docker** 2026 · [pjmitchell7.github.io/shadowprompt](https://pjmitchell7.github.io/shadowprompt)

- Architected an in-memory pre-inference proxy inspecting raw Unicode streams at 0.038ms P99 latency, blocking steganography and template breakouts.
- Integrated deceptive honeypot sandboxes seeded with canary credentials; generated compliant STIX 2.1 and ArcSight CEF syslog feeds for direct SIEM ingestion.

**PromptShield: Tokenization-Aware LLM Security Middleware | Python, FastAPI, scikit-learn** 2026

- Built asynchronous FastAPI middleware intercepting adversarial prompt injections with 0.27–1.11ms latency, avoiding expensive secondary LLM calls.
- Calibrated classification thresholds against federal compliance baselines, preventing evasion while maintaining high availability.

**Healthcare Network Traffic Anomaly Detection Framework | Python, scikit-learn** 2025

- Developed an automated network packet inspection pipeline featuring feature extraction and threshold calibration to support SOC operational alerting.
- Isolated anomalous traffic patterns using unsupervised ensemble detectors, validating performance under simulated network floods.